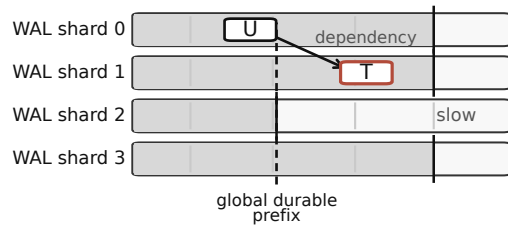


(a) Global durable prefix

Safe, but waits for unrelated slow shards

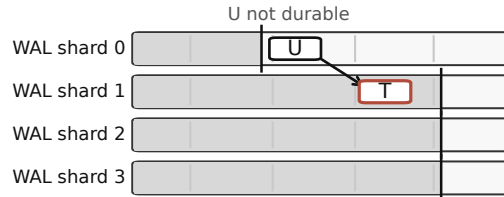


U and T are durable,
but T waits for slow shard

One slow shard holds back
transactions whose dependencies are durable

(b) Local-only acknowledgment

Fast, but unsafe under dependencies

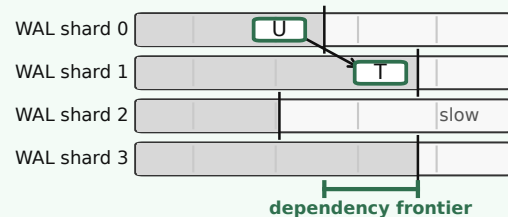


T is acked before
U is durable

Crash risk:
recover T without U

(c) Proposed method: TideWAL

Safe & selective: wait only for dependencies



Ack T after T and
its dependencies are durable

No global-prefix wait
for unrelated slow shards

■ durable region

| durable point

⋮ global prefix

□ log record